

MAScope: Diagnosing Collaboration Loss in Multi-Agent Systems

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ABSTRACT

Multi-agent systems assemble agents with different expertise, yet their performance depends on how that expertise is organized, not merely on how much of it is present. Existing benchmarks score the final outcome and therefore cannot tell what collaboration contributed to it. We introduce MAScope, an evaluation environment that builds 51 expert agents with private knowledge from Stack Exchange communities and provides 2,766 collaborative tasks whose required evidence and work dependencies are verifiable. MAScope traces every contribution from production to adoption and decomposes collaboration into four measurable capabilities: *expert selection*, *task orchestration*, *information transfer*, and *result integration*. Our experiments on twelve representative methods confirm that having the right experts does not deliver the task, with a 48% drop in success once work becomes interdependent. Analysis of complete collaboration trajectories attributes the loss to three gaps: (1) **planning rigidity** in task orchestration, (2) **semantic compression** in information transfer, and (3) **adjudicative aggregation** in result integration. Even exhaustive collaboration or the addition of memory modules cannot recover the loss. MAScope hence provides a common evaluation basis for measuring and improving multi-agent collaboration.

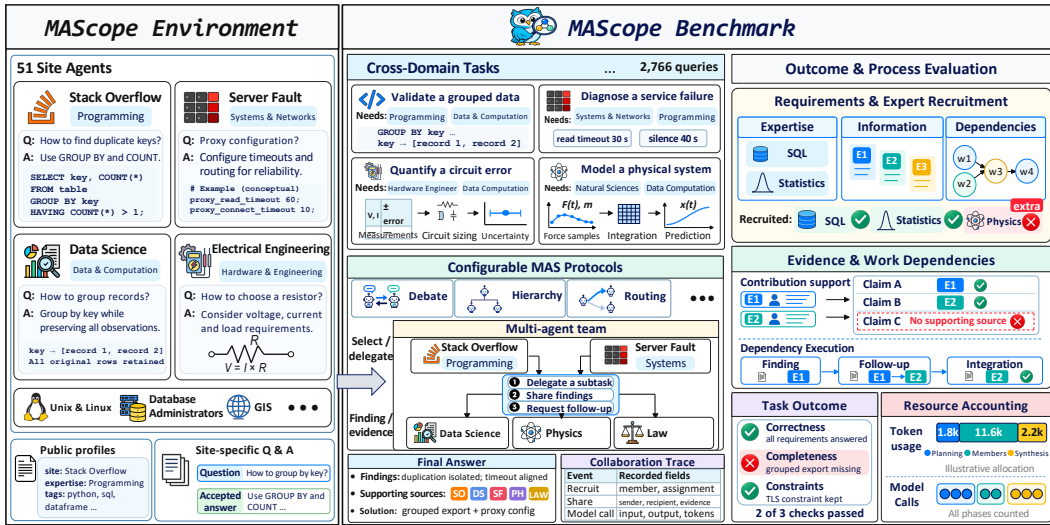


Figure 1: Overview of MAScope. The environment (left) provides 51 expert agents with public profiles and private Q&A corpora. The benchmark (right) lets methods recruit agents and collaborate autonomously on 2,766 cross-domain tasks, and verifies expert recruitment, evidence support, work dependencies, and task outcome while accounting for resource use.

1 INTRODUCTION

Agent tasks are moving from closed-form question answering toward interactive work in real environments such as software development, data analysis, and tool use (Trivedi et al., 2024). These

tasks demand cross-domain knowledge, long-horizon state tracking, and continual adaptation to feedback (Yao et al., 2024; Ma et al., 2024). Multi-agent systems (MAS) meet these demands by distributing a task among agents with different expertise and solving it through division of labor, communication, and mutual verification, and are regarded as a promising organizational form for complex tasks (Li et al., 2023a; Guo et al., 2024).

However, gathering agents does not by itself create collaborative value. A larger team brings in more expertise, but also more coordination overhead and more room for errors to propagate (Kim et al., 2025). Prior studies report that agents may misjudge their peers’ intentions (Li et al., 2023b), leave commitments unfulfilled and features in conflict in collaborative coding tasks (Khatua et al., 2026), and omit context, repeat reasoning, or aggregate prematurely in information-seeking tasks (Cemri et al., 2025). The same agents under different organizational structures also yield markedly different results (Zhuge et al., 2024; Qian et al., 2025). These observations suggest that the value of a team lies in the collaboration itself rather than in the abilities it assembles. The central challenge is to *identify what kind of agent collaboration is valuable and where it breaks down*.

Existing domain benchmarks (Jimenez et al., 2024; Mialon et al., 2024; Center for AI Safety et al., 2026) treat the whole system as a complete method and probe its capability limit through final scores (Kapoor et al., 2025), and therefore cannot distinguish why a MAS fails internally. Recent multi-agent benchmarks begin to measure the collaboration process through communication structure, coordination behavior, and framework-level capabilities (Zhu et al., 2025; Grötschla et al., 2025; Zhang et al., 2026; Orogat et al., 2026). These benchmarks, however, fix the team and the communication substrate in advance, showing that organization matters without locating where it fails. Our preliminary study (§ 2) shows that this blind spot is far from harmless: (1) the aggregation stage may compensate for insufficient agent work and credit collaboration that never happened, and (2) correct agent conclusions may degrade during interaction and be misattributed to agent ability. Attributing a MAS outcome to its collaboration therefore requires end-to-end evaluation across the whole workflow.

To address this gap, we introduce *MAScope*, an environment for collaborative problem solving by multiple experts. We build expert agents from the question-answer corpora of 51 Stack Exchange communities (Stack Exchange, Inc., 2014), one agent per community with a disjoint knowledge scope, so that every required contribution is indispensable and verifiable against its source. Within this environment, *MAScope* covers the three dependency relationships of real collaboration (*parallel*, *sequential*, and *clue-based expert discovery*) and provides 2,766 tasks from 499 task families, containing 11,018 required expertise units. *MAScope* monitors the complete collaboration process of a method and decomposes the task-execution workflow into four quantifiable capabilities for end-to-end evaluation: *expert selection*, *task orchestration*, *information transfer*, and *result integration*.

Experimental results on twelve influential MAS methods confirm that having experts present does not imply task completion. The twelve methods average only 41.3% success, and four of them are no better than a single agent reading the same corpora. Once work becomes interdependent rather than parallel, average success further drops from 57.8% to 29.8%. Our capability analysis attributes this failure to three gaps: (1) **planning rigidity** in task orchestration, where 20% of the follow-up work implied by predecessor findings is never initiated, (2) **semantic compression** in information transfer, where 20% of initiated work receives only a predecessor’s conclusion without the supporting information the next step needs, and (3) **adjudicative aggregation** in result integration, where 14% of correct local results are voted out instead of assembled into the answer. Comparing across coordination classes further shows that centralized methods are weakest in planning rigidity, semi-centralized methods in adjudicative aggregation, and decentralized methods in semantic compression.

Additionally, we force every method to continue collaborating until it confirms task completion, to test whether exhaustive collaboration yields better results. Success improves by at most 1.6% for any method, far below the 28-point dependency gap, and only methods that let findings re-enter planning benefit at all. A memory-method cross experiment then shows that no memory module consistently improves every host and that the best module differs across hosts. Our findings indicate that the bottleneck of multi-agent collaboration lies in the organizational mechanism rather than in the resources available to it. Improving collaboration therefore requires locating the specific gap in orchestration, transfer, or integration. *MAScope* makes these gaps measurable and comparable, providing a common basis for measuring and improving multi-agent collaboration.

2 MOTIVATIONS AND OBSERVATIONS

We first conduct an analysis to examine whether the final score of a MAS reflects the collaboration behind it. We run three representative methods, AutoGen (Wu et al., 2024), CrewAI (Moura, 2023), and Anemol (Ren et al., 2025), on multi-hop question answering (MuSiQue, 4-hop) and general assistant tasks (GAIA validation), with a shared backbone, four agents, and a budget of 150k tokens per task. For every run we freeze all agent records before aggregation and re-aggregate twice with the same rule, once from the complete record and once after masking the evidence contributed by one required agent. For every failed run we further check whether some agent output already contains a complete and correct answer. Settings and paired counts are given in Appendix F.

① **Final answers often stay correct after a required agent’s evidence is removed.** Masking one required agent lowers accuracy by only 12% on MuSiQue but by 48% on GAIA (Figure 2, left). The gap arises because MuSiQue evidence is textual and the aggregating agent can reason it out from the remaining context and its own knowledge, whereas GAIA evidence comes from file processing and web retrieval and cannot be reproduced by reasoning alone. A score therefore credits the team for work the aggregator did on its own, and the share of such credit varies severalfold across environments without ever showing in the score.

② **Correct agent answers are discarded before they reach the final answer.** Among failed runs, 20.1% on MuSiQue and 31.4% on GAIA already contain an agent output that fully answers the question with its key constraints (Figure 2, right). For AutoGen on GAIA this holds for 34 of 106 failures. In these runs the team has done the work and then lost it during handoff or aggregation, and no improvement to individual agents would recover it. The score nevertheless charges these losses to agent ability.

Together, the two observations show that an outcome score cannot separate obtaining a contribution from using it. Evaluating collaboration therefore requires every required contribution to be indispensable, ruling out compensation, and traceable from production to adoption, exposing where it is lost.

3 MASCOPE: TRACING COLLABORATION FROM PRODUCTION TO ADOPTION

To evaluate multi-agent collaboration end to end, we develop MAScope, an environment where every required contribution is indispensable and can be followed from the agent that produces it to the answer that adopts it. First, we build an expert environment from 51 Stack Exchange communities and compose collaborative tasks on top of it (§ 3.1). Second, we organize these tasks along two independent dimensions, how their pieces of work depend on each other and how much expertise they require (§ 3.2). Third, we design an evaluation framework that follows each contribution through the collaboration workflow and measures four capabilities (§ 3.3).

3.1 BENCHMARK CONSTRUCTION

Expert environment. We instantiate one expert agent per Stack Exchange community, covering 51 communities in total. An agent is defined by a public profile and a private corpus. Methods see only the profile, which lists the site name, expertise scope, and topic tags, when deciding whom to recruit. Execution instead draws on the corpus, which holds the community’s questions, accepted answers, and source identifiers for retrieval and citation. Disjoint corpora prevent one agent from

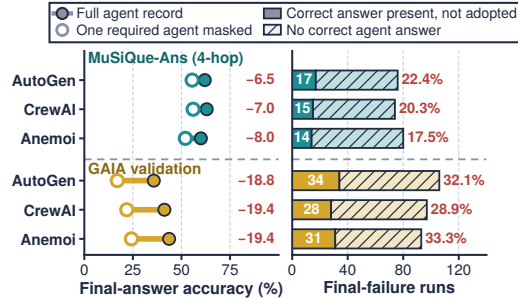


Figure 2: MAS collaboration results on MuSiQue and GAIA. Left, final-answer accuracy of three methods with the full agent record and with one required agent’s evidence masked. Right, failed runs split by whether a correct agent answer was already present but not adopted.

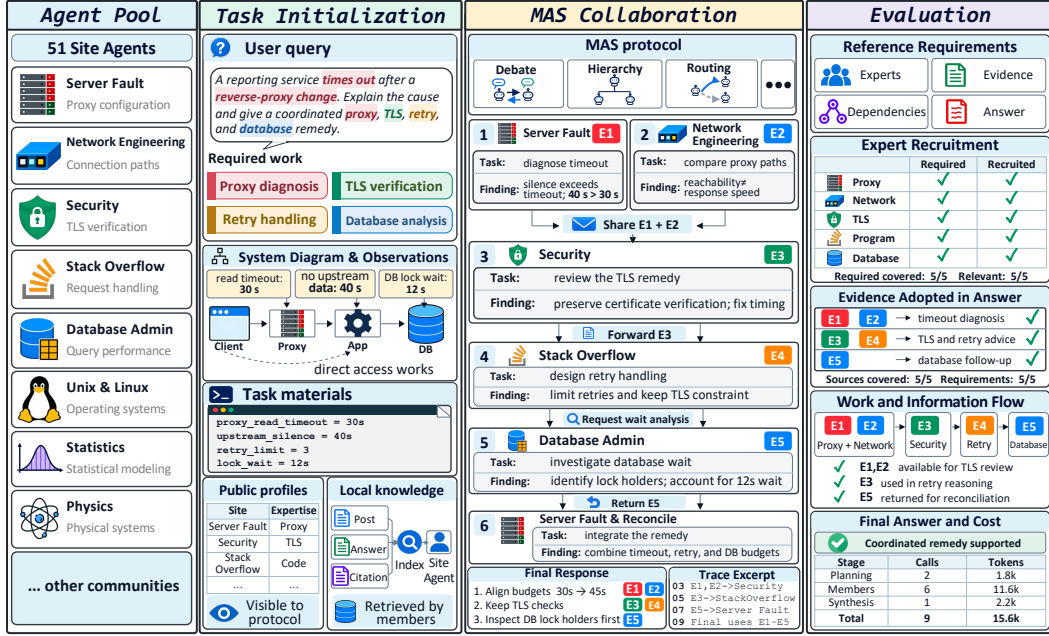


Figure 3: A MAScope run on a service-diagnosis task. The task requires five information units, of which E3 to E5 depend on the diagnosis carried in E1 and E2. The evaluator checks which experts were recruited, whether each dependency was carried forward with its constraints, which evidence reached the final answer, and how many tokens each stage consumed.

supplying evidence that belongs to another, while source identifiers internal to the benchmark keep a memorized copy of Stack Exchange from passing as evidence. Since agents share one backbone and differ only in their corpora, task completion turns on how their information is organized rather than on any agent being stronger.

Task definition. A task is a tuple $T = (q, U, G, V)$. The method sees only the question q , which describes a user’s goal and observations without naming any site. Behind it, U enumerates the information units a solution requires, each bound to a source identifier in one corpus, G records which of those units can be scoped only once a predecessor finding is in hand, and V states the acceptance criteria for the final answer. Tasks admit multiple valid paths, and verified equivalent evidence and alternative routes are recorded in the reference, so evaluation does not presume a fixed order of agent calls.

Construction pipeline. On top of this environment, each task is built in four steps. **I. Evidence alignment:** from real community questions and answers, we select information units that belong to different sites yet serve one user goal, verify that each is supported by its corpus, and bind it to a source identifier. **II. Task composition:** the demands of these units are rewritten into a single user-facing question that gives the goal and the observed symptoms while leaving the required expertise to be inferred from them. **III. Dependency annotation:** a unit is marked dependent when its inputs or objectives can be fixed only from a predecessor finding. We confirm each dependency by solving the unit locally with and without that finding, and record an expert-discovery relationship when the predecessor finding is also what reveals the expertise needed next. **IV. Validation and freezing:** to ensure that a task is well posed, we solve it end to end, confirming that every corpus is reachable and the required expertise is inferable from q . Each unit is also answered against its own corpus alone, and units whose evidence cannot be recovered this way are revised or dropped. Tasks are then frozen by family, with variants of one family kept in the same evaluation split. The release package contains q , the profiles, and the environment, while U , G , and V remain with the evaluator.

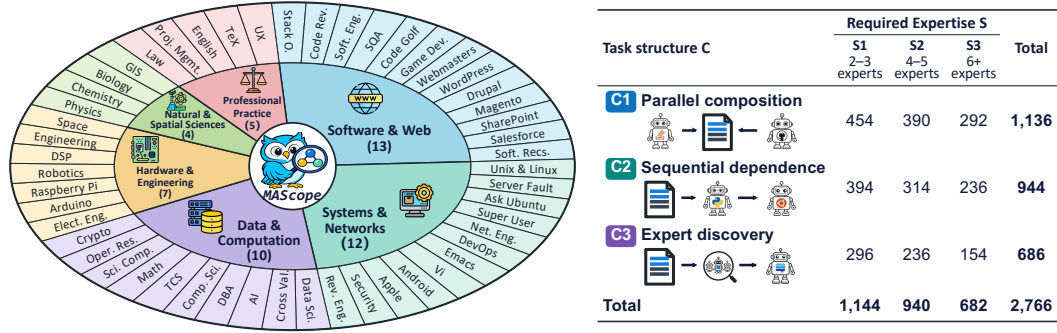


Figure 4: Expert pool and task taxonomy. Left, 51 expert agents in six resource groups. Right, number of queries in each cell of collaboration structure (C1 to C3) by required expertise (S1 to S3).

3.2 BENCHMARK OVERVIEW

Statistics. MAScope contains 2,766 queries from 499 task families with 11,018 required expertise units. The 51 agents span six resource groups, from software engineering to professional practice, and a single task usually draws on more than one group, producing integrated goals in system diagnosis, data processing, engineering analysis, and professional judgment.

Task taxonomy. We characterize each task by its collaboration structure C and its required expertise S . *Parallel composition* (C1) splits a task into independent pieces that are combined at the end, such as a data transformation rule together with a check for its result. *Sequential dependence* (C2) chains them instead, so that a later piece takes an earlier finding as its input or objective, as when the choice of a configuration change follows from where a timeout originates. In *expert discovery* (C3), even the identity of the next expert is hidden in a predecessor finding, and the method has to recognize it while solving. S counts the professional roles a task requires, S1 (2–3 experts), S2 (4–5 experts), and S3 (6–8 experts), and places no limit on how many agents a method may call. With C and S varying independently, MAScope isolates the effect of deepening dependency from that of widening expertise on the same method.

3.3 EVALUATION FRAMEWORK

Autonomous execution and outcome verification. A method receives q , the public profiles, a shared agent pool, tools, and a budget, and then recruits agents, assigns work, exchanges information, and submits a final answer entirely under its own mechanism, while the environment logs every agent call, message, local output, and token. The evaluator verifies the submitted answer against V to obtain task success, maps the evidence cited in the answer onto U to obtain evidence coverage, which records how much of the required content was completed even when the task fails, and sums tokens across planning, execution, aggregation, and memory operations to obtain execution cost.

Tracing contributions along the dependency chain. To locate where collaboration breaks down, we follow each required contribution along the chain that should carry it into the answer. **Expert selection** comes first and is measured over all queries by the recall and precision of the participating agents A_q against the required experts R_q , which separates omitted expertise from irrelevant recruitment. For each C2 and C3 query we then preregister one critical dependency, an edge of G that every valid solution must traverse before reaching the answer, and a method is ready to act on it once it has obtained the predecessor finding and recruited the downstream expert ($s^{(1)}$). Assigning the follow-up work to that expert marks **task orchestration** ($s^{(2)}$), and carrying the predecessor’s evidence with its key constraints into that assignment marks **information transfer** ($s^{(3)}$). The expert then produces a correct local result ($s^{(4)}$), whose survival into the final answer marks **result integration** ($s^{(5)}$). Writing $s_q^{(k)} \in \{0, 1\}$ for whether stage k is reached, conditional on stage $k - 1$,

$$\mathcal{P} = \frac{\sum_q s_q^{(2)}}{\sum_q s_q^{(1)}}, \quad \mathcal{C} = \frac{\sum_q s_q^{(3)}}{\sum_q s_q^{(2)}}, \quad \mathcal{L} = \frac{\sum_q s_q^{(4)}}{\sum_q s_q^{(3)}}, \quad \mathcal{I} = \frac{\sum_q s_q^{(5)}}{\sum_q s_q^{(4)}},$$

Table 1: Main results on MAScope. Task success (%) across nine task cells, crossing collaboration structure, namely **C1** parallel composition, **C2** sequential dependence, and **C3** expert discovery, with required expertise S1 to S3. Shading marks degradation from the C1 cell at the same scale. Overall gives success on all 2,766 queries (SR) and its relative change from C1 to C2UC3 (Δ_{dep}) and from S1 to S3 (Δ_{scale}), where a smaller loss is better. Cost is mean tokens (k) and model calls per query, and Avg. is the class mean. Values are means over five runs, with the best and second-best MAS method in **bold** and underlined.

Method	C1 Parallel composition			C2 Sequential dependence			C3 Expert discovery			Overall			Cost↓		
	S1	S2	S3	S1	S2	S3	S1	S2	S3	SR↑	Δ _{dep}	Δ _{scale}	Tok. (k)	Calls	
Centralized	AutoGen	74.3	65.7	53.9	57.6	44.0	28.1	49.6	34.4	20.6	52.1	−35.9	−39.7	171	14.6
	Magentic-One	72.5	64.9	52.7	53.4	41.0	24.7	48.4	33.5	21.0	50.2	−38.3	−39.9	208	23.2
	Flow	75.0	65.8	53.3	45.5	34.0	20.5	38.7	26.1	16.5	46.6	−50.4	−39.4	126	14.4
	CrewAI	77.2	68.9	55.3	43.2	27.0	15.8	33.6	21.1	12.1	44.9	−58.9	−41.1	119	11.8
	LangGraph	64.8	54.2	41.2	47.9	33.5	22.6	36.1	27.4	15.8	42.3	−39.5	−43.6	127	16.7
Semi-centralized	Avg.	72.8	63.9	51.3	49.5	35.9	22.4	41.3	28.5	17.2	47.2	−44.6	−40.8	150	16.1
	Anemoui	64.3	57.4	54.2	42.6	33.4	25.3	39.3	31.3	17.8	44.3	−43.1	−28.5	118	15.5
	AgentVerse	67.6	60.2	48.8	39.8	27.3	16.4	37.7	25.3	15.1	41.9	−51.5	−40.4	165	19.9
	MoA	61.2	55.1	44.9	21.2	12.4	6.6	25.6	13.6	9.2	32.0	−70.9	−38.2	1,044	94.6
	MAD	52.2	42.6	30.9	25.9	18.7	11.2	26.9	17.7	8.2	29.4	−54.6	−48.2	125	16.2
Decentralized	Avg.	61.3	53.8	44.7	32.4	23.0	14.9	32.4	22.0	12.6	36.9	−55.0	−38.8	363	36.6
	GoA	65.6	62.9	60.5	43.3	36.3	29.4	36.4	27.5	19.7	46.2	−46.1	−19.5	208	26.8
	AgentNet	56.7	43.5	29.2	30.8	20.8	13.2	39.9	27.3	13.6	33.7	−42.7	−53.5	88	7.9
	SelfOrg	50.5	43.5	42.7	29.4	19.4	19.2	27.0	20.5	11.8	32.3	−51.0	−25.8	324	104.3
	Avg.	57.6	50.0	44.1	34.5	25.5	20.6	34.4	25.1	15.1	37.4	−46.6	−33.0	206	46.3
Single Agent	54.7	43.2	32.3	32.9	23.6	14.8	28.9	19.6	13.8	32.6	−46.5	−45.6	131	11.9	

where $\mathcal{L}$ serves as an agent-side reference that tells an agent failing to solve apart from a team failing to adopt, and the product $\mathcal{PCLI}$ gives dependency survival, the share of ready dependencies that reach the answer. All stages are decided from the logged trajectory by matching source identifiers and preregistered terms, keeping every capability comparable across coordination structures.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Compared methods. We evaluate twelve MAS methods grouped by how centralized their coordination is. *Centralized* methods route every decision through one coordinator: AutoGen, LangGraph, CrewAI, Magentic-One (Fourney et al., 2024), and Flow (Niu et al., 2025). *Semi-centralized* methods keep a planning, aggregation, or evaluation role while letting agents answer in parallel or talk directly: Anemoui (Ren et al., 2025), AgentVerse (Chen et al., 2024), MoA (Wang et al., 2025), and MAD (Du et al., 2023). *Decentralized* methods have no fixed coordinator and let agents propagate information or route work themselves: GoA (Yun et al., 2026), AgentNet (Yang et al., 2025), and SelfOrg (Tastan et al., 2026). Each runs under its native mechanism and decides for itself whom to recruit, how to organize the work, and when to stop. As a reference we add a single agent with access to all 51 corpora, which holds the whole task in one execution context.

Implementation details. Every method and agent uses `claude-sonnet-5` (Anthropic, 2026) as its backbone and shares the same expert agents, profiles, and retrieval interface. Each task is capped at 2M tokens and 256 model calls, and tables report actual consumption. We run every method five times over all 2,766 queries and report the mean. Details are in Appendix C.

4.2 MAIN RESULTS

① **A MAS does not reliably outperform a single agent.** Two thirds of the methods improve on the single agent’s 32.6% by at least 28%, led by AutoGen at 52.1%. The other four stay within 10%

Table 2: Collaboration capabilities along the dependency chain (%). Expert selection reports recall and precision over all queries. The four ► columns are conditional rates over the 1,630 preregistered dependencies, each conditioned on the stage before it, where local solve is an agent-side reference giving correct local results per usable input. Dep. survival is their product, and evidence coverage is the query mean of required information supported in the final answer. Shading marks loss in selection and the three collaboration stages. Values are means over five runs, with the best and second-best in **bold** and underlined.

Method	🧑‍🔬 Expert selection		🎛️ Task orchestr.	➡️ 🔄 Info. transfer	🔧 Local solve	➡️ 🧩 Result integr.	🛑 Dep. survival	📁 Evidence coverage	
	Recall	Precision							
Centralized	AutoGen	81.8	<u>74.0</u>	85.5	89.8	<u>91.6</u>	<u>92.3</u>	64.9	<u>67.4</u>
	Magentic-One	80.4	71.6	86.5	83.5	91.3	92.2	<u>60.8</u>	65.7
	Flow	83.1	71.9	83.2	<u>85.4</u>	90.4	90.1	57.9	63.6
	CrewAI	81.9	73.7	<u>70.9</u>	81.8	89.4	91.9	47.6	61.8
	LangGraph	<u>77.7</u>	<u>68.6</u>	<u>77.3</u>	<u>85.3</u>	<u>89.7</u>	94.4	55.8	58.1
	Avg.	<u>81.0</u>	<u>72.0</u>	<u>80.7</u>	<u>85.2</u>	<u>90.5</u>	<u>92.2</u>	<u>57.4</u>	<u>63.3</u>
Semi-centralized	Anemoui	78.9	69.4	92.0	80.5	93.1	76.4	52.6	58.5
	AgentVerse	<u>86.6</u>	66.7	76.0	77.0	90.6	79.8	42.3	58.7
	MoA	100.0	7.8	<u>34.8</u>	77.7	90.4	80.0	19.6	72.1
	MAD	<u>65.1</u>	70.0	87.8	83.3	87.6	<u>73.9</u>	47.3	44.0
	Avg.	<u>82.6</u>	<u>53.5</u>	<u>72.6</u>	<u>79.6</u>	<u>90.4</u>	<u>77.5</u>	<u>40.5</u>	<u>58.3</u>
Decentralized	GoA	84.1	<u>69.6</u>	<u>91.7</u>	<u>71.7</u>	88.4	87.3	50.8	62.6
	AgentNet	<u>68.1</u>	87.0	87.6	79.3	90.5	85.9	54.0	49.6
	SelfOrg	100.0	<u>7.8</u>	85.5	<u>65.6</u>	86.9	81.2	39.6	49.2
	Avg.	<u>84.1</u>	<u>54.8</u>	<u>88.3</u>	<u>72.2</u>	<u>88.6</u>	<u>84.8</u>	<u>48.1</u>	<u>53.8</u>

of it, and MoA, MAD, and SelfOrg even fall below, though MoA and SelfOrg put all 51 experts to work. With corpora, budget, and backbone held fixed, a MAS is separated from the single agent only by how it organizes the work after dividing it.

② **Complex work dependencies systematically degrade MAS performance.** When a MAS takes on tasks whose later work builds on earlier findings, average success falls from 57.8% to 29.8%. Every coordination class loses up to 55% of its success, so no organizational style is immune. Scaling the required expertise from 2–3 to 6+ experts cuts success by a further 38%. MoA and SelfOrg activate the entire pool and never miss a required expert, yet both trail the single agent on sequential-dependence tasks. The loss thus falls on the step that turns one agent’s finding into the next agent’s work, not on the availability of expertise.

③ **Higher cost does not translate into better performance.** Per-task cost spans an order of magnitude, from 88k tokens for AgentNet to 1,044k for MoA, and tracks how many agents answer at once rather than how well they answer. Success, however, does not track cost at all. Flow reaches 46.6% at 126k tokens while MoA reaches 32.0% at 8× the cost, and AgentNet, the cheapest method of all, still edges past the single agent. The performance–cost frontier belongs to the centralized class, where AutoGen is highest at 1.3× the single agent’s cost and Flow and CrewAI beat it at comparable cost. Extra computation therefore buys wider participation and communication rather than the piece of work that is missing.

4.3 WHY DOES COLLABORATION FAIL?

To locate where collaboration breaks down, we trace every preregistered dependency through the four capabilities of § 3.3. The results indicate that agents are reliable at completing work on their own, and hence reveal three capability gaps in the MAS collaboration process.

① **Agent findings struggle to revise an established plan.** In 20% of dependencies a finding arrives but the work it implies is never assigned, and we call this **planning rigidity** in task orchestra-

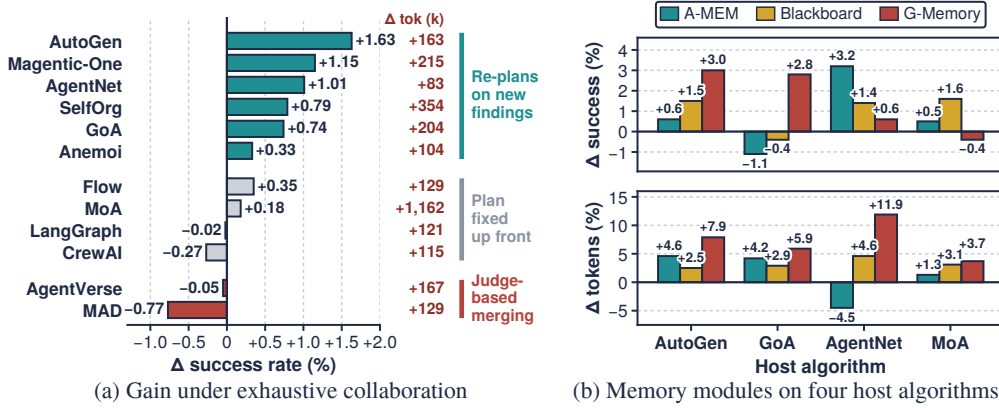


Figure 5: Exhaustive collaboration and memory modules. (a) Change in success when a method must keep collaborating until it confirms completion, relative to its own termination, grouped by whether findings re-enter planning, the plan is fixed up front, or results are merged by adjudication. Red figures give the extra tokens each method spends. (b) Change in success and tokens when three memory modules are added to four host methods. Values are means over five runs.

tion. Centralized methods follow a decomposition-before-execution pipeline, so a finding produced at run time has no route back into the plan. CrewAI executes only predefined tasks and assigns 70.9% of the work its findings imply, and LangGraph 77.3%. Methods with a replanning trigger fare no better, as Magentic-One and Flow revise only when progress stalls or a step fails, reaching 86.5% and 83.2%. Orchestration holds up where an agent can claim work for itself, with Anemoi, GoA, and AgentNet between 87.6% and 92.0%. MoA is an extreme case at 34.8%, aggregating one round of parallel answers without assigning any follow-up work.

② **Agents pass on conclusions without their supporting evidence.** Of the work that is assigned, another 20% arrives as a bare conclusion stripped of the evidence and constraints behind it, and we call this **semantic compression** in information transfer. Decentralized methods relay each finding hop by hop, and a restatement cannot carry what the next step has not asked for yet. SelfOrg and GoA broadcast round summaries and deliver usable inputs in 65.6% and 71.7% of cases, while AgentVerse, which reduces an assignment to a single subtask sentence, reaches 77.0%. Transfer holds up wherever the predecessor’s output is passed on intact, whether a coordinator forwards the full conversation as in AutoGen at 89.8%, or a graph edge carries the output itself as in Flow at 85.4%.

③ **Correct local results are overridden during aggregation.** Once a correct local result exists, 14.5% of the time it still never reaches the answer, and we call this **adjudicative aggregation** in result integration. Semi-centralized methods treat agent outputs as competing answers and select among them, a rule that rewards agreement over correctness and discards a correct result held by a single agent. MAD retains 73.9% and Anemoi 76.4%, and moving the decision to a different role does not change it. Centralized methods assemble outputs into one answer instead of choosing between them, and all retain more than 90%.

Overall, each MAS class carries its own strength and its own blind spot. A central coordinator holds the full context and assembles the final answer, which secures transfer and integration at the cost of a plan fixed before any finding exists. Decentralized methods remove that coordinator and let any agent pick up new work, but the hop-by-hop relay that enables it thins out the evidence along the way. Semi-centralized methods sit between the two, buying parallel progress with an arbitration step that can overwrite a result already correct.

4.4 CAN EXHAUSTIVE COLLABORATION RECOVER PERFORMANCE?

To test whether more collaboration recovers the capability gaps of a MAS, we force each method to keep going under its own mechanism until it confirms that every requirement is met and no agent can make any further contribution.

① **Exhaustive collaboration does not recover MAS performance.** Collaborating to exhaustion improves success by at most 1.6% and lowers it for four methods, while every method spends about twice the tokens it did before. Set against the 28-point gap between parallel and dependency tasks, even the best of these gains recovers under 6% of it. The extra rounds go into re-running the same plan, regenerating the same answer, or adjudicating again, so work a method never arranged before it would have stopped remains unarranged afterwards.

② **Only methods that turn new findings into new work benefit.** Every method that lets findings re-enter planning gains, from 0.33% for Anemio to 1.63% for AutoGen, because continuing means revisiting what has been found and dispatching what follows from it. Methods whose plan is fixed up front stay between -0.27% and $+0.35\%$, since another round executes the same plan and any gain is offset by fresh errors. Methods that merge by adjudication decline, with MAD losing 0.77% , as every additional round is another chance to overwrite a result that was already correct. The methods carrying the largest orchestration gap are therefore the ones least able to use the extra collaboration.

4.5 COMBINING MEMORY MODULES WITH MAS METHODS

Memory supplies the experience a MAS can reuse when selecting experts, making later decisions, and reusing information, which makes it another common enhancement. We adopt three influential modules into MAScope to quantify the value of combining a MAS with memory. A-MEM (Xu et al., 2025) keeps cross-task experience as structured notes on each agent, Blackboard (Salemi et al., 2025) holds a shared working state that the team reads and writes within a task, and G-Memory (Zhang et al., 2025a) retrieves across tasks and steps through a three-level graph of experience, tasks, and interactions. We pair each module with AutoGen, GoA, AgentNet, and MoA, four representative methods drawn from the three coordination classes.

① **A memory module helps only where it supplies what the host lacks.** Every module lowers success on at least one host, and the best module differs from host to host. AgentNet routes work without knowing what individual agents do well, which A-MEM’s agent notes supply, for the largest gain of 3.2% . G-Memory’s cross-step retrieval addresses the dependent work that AutoGen and GoA fail to connect, giving 3.0% and 2.8% , and Blackboard’s shared state lets MoA agents see one another’s work, worth 1.6% . The reverse pairings fail. Cross-step retrieval costs MoA 0.4% because MoA does no cross-step work, while agent notes and shared state cost GoA 1.1% and 0.4% because its losses come from relaying summaries.

② **Memory gains are usually paid for in tokens.** Retrieval and writing add 1% to 12% to consumption across most pairings, and the additions are largest where the gains are smallest, as with G-Memory on AgentNet at 12% more tokens for 0.6% more success. AgentNet with A-MEM is the one pairing that pays for itself, cutting tokens by 4.5% while raising success by 3.2% , because knowing what each agent is good at removes the routing attempts that lead nowhere. Experience that enters a decision saves the trial and error it replaces, and experience that does not leaves only the cost of retrieving it.

5 RELATED WORK

MAS methods. MAS methods differ mainly in who decides how work is organized. Centralized methods route that decision through a coordinator, evolving from fixed roles and standard operating procedures to speaker selection, shared-state routing, and dispatch over task ledgers and subtask graphs (Hong et al., 2024; Qian et al., 2024; Wu et al., 2024; Barda, 2025; Moura, 2023; Fournery et al., 2024; Niu et al., 2025). Semi-centralized methods keep a planning, aggregation, or evaluation role while agents answer in parallel or talk to each other directly (Chen et al., 2024; Du et al., 2023; Wang et al., 2025; Ren et al., 2025). Decentralized methods drop the coordinator and let agents propagate information or route work themselves (Zhuge et al., 2024; Liu et al., 2024b; Qian et al., 2025; Yun et al., 2026; Tastan et al., 2026; Yang et al., 2025), and a further line searches for the organization automatically (Zhang et al., 2025c; Hu et al., 2025; Zhang et al., 2025b). Such diversity leaves each mechanism buried under the others in a single final score. MAScope therefore evaluates all of them in one expert environment and attributes their differences stage by stage.

Evaluation of multi-agent collaboration. General agent benchmarks score a system by what it finally completes (Liu et al., 2024a; Mialon et al., 2024; Jimenez et al., 2024; Wu et al., 2026). Multi-agent benchmarks open up the process, measuring milestones and communication structure (Zhu et al., 2025), distributed reasoning on prescribed topologies (Grötschla et al., 2025), coordination under information silos (Zhang et al., 2026), commitments and conflicts in collaborative development (Khatua et al., 2026), framework-level capabilities (Orogat et al., 2026), and teamwork under enforced role separation (Kim et al., 2026), while a parallel line reads trajectories for failure taxonomies (Cemri et al., 2025), scaling conditions (Kim et al., 2025), and agent credit (Jiang et al., 2026). These benchmarks each measure collaboration from a single facet, and none follows a required contribution from production to adoption across the whole workflow. MAScope makes each contribution indispensable through private corpora and its use verifiable through hidden references, resolving an outcome difference into four comparable capabilities.

6 CONCLUSION

We introduce MAScope, an environment that builds 51 expert agents with disjoint private knowledge and traces every required contribution from the agent that produces it to the answer that adopts it. Experiments on twelve influential MAS methods show that having the required experts present does not deliver the task. Success halves once work becomes interdependent, and three gaps account for it: planning rigidity in task orchestration, semantic compression in information transfer, and adjudicative aggregation in result integration. Even exhaustive collaboration or the addition of memory modules cannot recover the loss. MAScope turns these gaps into verifiable steps, so that a proposed improvement can be located and tested stage by stage.

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